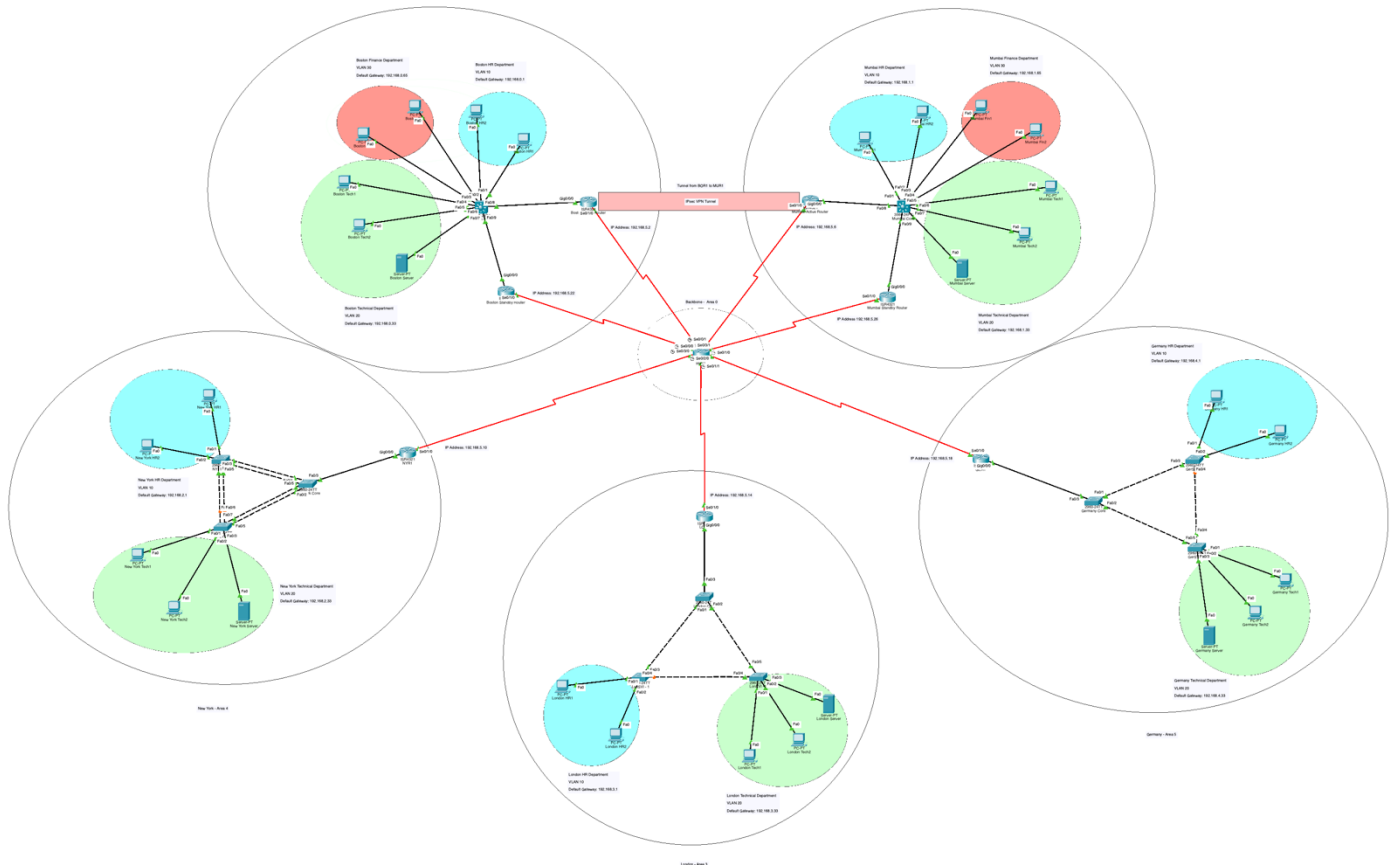


Secure Enterprise Network Design

I. Project Design

Topology of the Multi-Location Inter-Networking Strategy for a Global Organization



II. Network Description:

Boston Office (Area 1) - Head Quarters

Devices:

- 6 PCs (Boston Tech1, Boston Tech2, Boston HR1, Boston HR2, Boston Fin1, Boston Fin2)
- 1 Server (Boston DHCP Server),
- 1 L-3 core switch (Boston Core)
- 1 Router (BOR1)

Total Devices: 9

- **Departments:** HR (VLAN 10), Technical (VLAN 20), Finance (VLAN 30).
- **IP Ranges:**
 - HR: 192.168.0.0/26 (Def Gateway: 192.168.0.1)
 - Technical: 192.168.0.32/26 (Def Gateway: 192.168.0.33)
 - Finance: 192.168.0.64/26 (Def Gateway: 192.168.0.65)

Boston DHCP Configuration:

Boston Server

Physical Config Services **Desktop** Programming Attributes

IP Configuration [X]

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.0.34

Subnet Mask: 255.255.255.224

Default Gateway: 192.168.0.33

DNS Server: 192.168.0.34

Boston Server

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

Start IP Address : 0 0 0 0

Subnet Mask: 0 0 0 0

Maximum Number of Users : 0

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	
serverPool	0.0.0.0	0.0.0.0	0.0.0.0	0.0.0.0	0	0
Boston Finance Department	192.168.0.65	192.168.0.34	192.168.0.66	255.255.255.224	30	0
Boston Technical Department	192.168.0.33	192.168.0.34	192.168.0.35	255.255.255.224	29	0
Boston HR Department	192.168.0.1	192.168.0.34	192.168.0.2	255.255.255.224	30	0

Boston DNS Configuration:

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DNS

DNS Service

On

Off

Resource Records

Name

Type

A Record

Address

Add

Save

Remove

No.	Name	Type	Detail
0	BBR1	A Record	192.168.5.1
1	BBR1	A Record	192.168.5.5
2	BBR1	A Record	192.168.5.9
3	BBR1	A Record	192.168.5.13
4	BBR1	A Record	192.168.5.17
5	BBR1	A Record	192.168.5.21
6	BBR1	A Record	192.168.5.25
7	BOR1	A Record	192.168.5.2
8	BOR2	A Record	192.168.5.22
9	GER1	A Record	192.168.5.18
10	LOR1	A Record	192.168.5.14
11	MUR1	A Record	192.168.5.6
12	MUR2	A Record	192.168.5.26
13	NYR1	A Record	192.168.5.10

Boston VLAN Configuration:

```
Boston-Core#
Boston-Core#sh vlan b
```

VLAN	Name	Status	Ports
1	default	active	Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Boston_HR_Department	active	Fa0/1, Fa0/2
20	Boston_Technical_Department	active	Fa0/5, Fa0/6, Fa0/7
30	Boston_Finance_Department	active	Fa0/3, Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Mumbai Office (Area 2) - Head Quarters

Devices:

- 6 PCs (Mumbai Tech1, Mumbai Tech2, Mumbai HR1, Mumbai HR2, Mumbai Fin1, Mumbai Fin2)
- 1 server (Mumbai DHCP Server)
- 1 L-3 core switch (Mumbai Core)
- 1 router (MUR1)

Total Devices: 9

- **Departments:** HR (VLAN 10), Technical (VLAN 20), Finance (VLAN 30).
- **IP Ranges:**
 - HR: 192.168.1.0/26 (Def Gateway: 192.168.1.1)
 - Technical: 192.168.1.32/26 (Def Gateway: 192.168.1.33)
 - Finance: 192.168.1.64/26 (Def Gateway: 192.168.1.65)

Boston DHCP Configuration:

The screenshot shows a window titled "Mumbai Server" with a tabbed interface. The "Desktop" tab is selected. Inside the Desktop tab, there is a sub-tabbed interface with "IP Configuration" selected. The "IP Configuration" sub-tab shows two radio buttons: "DHCP" (unselected) and "Static" (selected). Below the radio buttons, there are four input fields: "IPv4 Address" with the value "192.168.1.34", "Subnet Mask" with the value "255.255.255.224", "Default Gateway" with the value "192.168.1.33", and "DNS Server" with the value "0.0.0.0".

Field	Value
IPv4 Address	192.168.1.34
Subnet Mask	255.255.255.224
Default Gateway	192.168.1.33
DNS Server	0.0.0.0

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface

FastEthernet0

Service

On

Off

Pool Name

serverPool

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

Start IP Address :

0

0

0

0

Subnet Mask:

0

0

0

0

Maximum Number of Users :

0

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User
serverPool	0.0.0.0	0.0.0.0	0.0.0.0	0.0.0.0	0
Mumbai HR Department	192.168.1.1	192.168.0.34	192.168.1.2	255.255.255.224	30
Mumbai Technical Department	192.168.1.33	192.168.0.34	192.168.1.35	255.255.255.224	29
Mumbai Finance Department	192.168.1.65	192.168.0.34	192.168.1.66	255.255.255.224	30

Mumbai VLAN Configuration:

```
Mumbai-Core#
Mumbai-Core#sh vlan b
```

VLAN	Name	Status	Ports
1	default	active	Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	Mumbai_HR_Department	active	Fa0/1, Fa0/2
20	Mumbai_Technical_Department	active	Fa0/5, Fa0/6, Fa0/7
30	Mumbai_Finance_Department	active	Fa0/3, Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

New York Office (Area 4)

Devices:

- 4 PCs (New York Tech1, New York Tech2, New York HR1, New York HR2)
- 1 server (New York DHCP Server),
- 3 L-2 switches (NewYork-Core, NY-10, NY-20)
- 1 Router (NYR1)

Total Devices: 8

- **Departments:** HR (VLAN 10), Technical (VLAN 20).
- **IP Ranges:**
 - HR: 192.168.2.0/26 (Def Gateway: 192.168.2.1)
 - Technical: 192.168.2.32/26 (Def Gateway: 192.168.2.33)

New York DHCP Configuration:

The screenshot shows a web-based configuration interface for a server named "New York Server". The interface has a top navigation bar with tabs: Physical, Config, Services, Desktop (selected), Programming, and Attributes. Below the navigation bar, there is a blue header for the "IP Configuration" section. The main content area shows the "IP Configuration" settings. The "Static" radio button is selected, and the "DHCP" radio button is unselected. The fields for IP Configuration are as follows:

Field	Value
IPv4 Address	192.168.2.34
Subnet Mask	255.255.255.224
Default Gateway	192.168.2.33
DNS Server	0.0.0.0

New York Core VLAN Configuration:

New York HR Side VLAN Configuration:

```
NY-10#  
NY-10#sh vlan b
```

VLAN	Name	Status	Ports
1	default	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	NY_HR_Department	active	Fa0/1, Fa0/2
20	NY_Technical_Department	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

New York Technical Side VLAN Configuration:

```
NY-20#  
NY-20#show vlan b
```

VLAN	Name	Status	Ports
1	default	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	NY_HR_Department	active	
20	NY_Technical_Department	active	Fa0/1, Fa0/2, Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

London Office (Area 3)

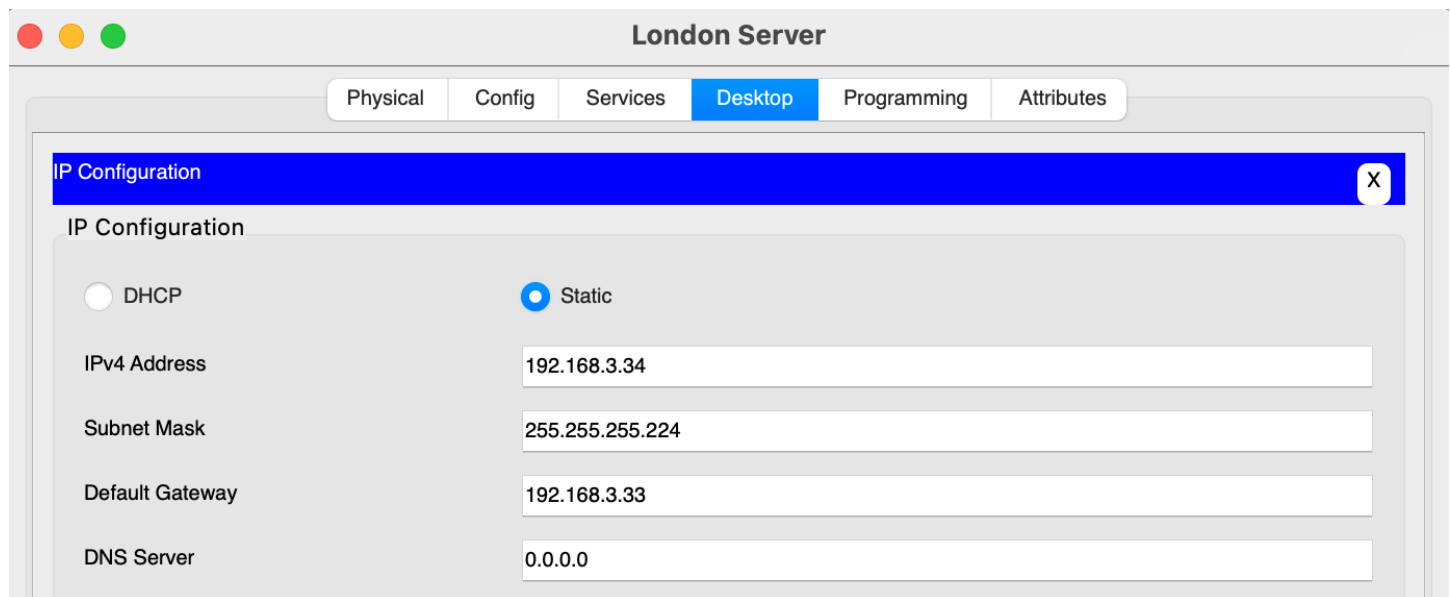
Devices:

- 4 PCs (London Tech1, London Tech2, London HR1, London HR2)
- 1 server (London DHCP Server),
- 3 L-2 switches (London -Core, LonSVI-1, LonSVI-2)
- 1 Router (LOR1)

Total Devices: 8

- **Departments:** HR (VLAN 10), Technical (VLAN 20).
- **IP Ranges:**
 - HR: 192.168.3.0/26 (Def Gateway: 192.168.3.1)
 - Technical: 192.168.3.32/26 (Def Gateway: 192.168.3.33)

London DHCP Configuration:



The screenshot shows a web-based configuration interface for a "London Server". At the top, there are tabs for "Physical", "Config", "Services", "Desktop" (which is selected and highlighted in blue), "Programming", and "Attributes". Below the tabs, there is a section titled "IP Configuration" with a blue header bar and a close button (X). Inside this section, there are two radio buttons: "DHCP" (unselected) and "Static" (selected). Below the radio buttons, there are four input fields for static IP configuration: "IPv4 Address" (192.168.3.34), "Subnet Mask" (255.255.255.224), "Default Gateway" (192.168.3.33), and "DNS Server" (0.0.0.0).

Field	Value
IPv4 Address	192.168.3.34
Subnet Mask	255.255.255.224
Default Gateway	192.168.3.33
DNS Server	0.0.0.0

Physical
Config
Services
Desktop
Programming
Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface

FastEthernet0 v

Service ☒ On ☐ Off

Pool Name

serverPool

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

Start IP Address :

0

0

0

0

Subnet Mask:

0

0

0

0

Maximum Number of Users :

0

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User
serverPool	0.0.0.0	0.0.0.0	0.0.0.0	0.0.0.0	0
London Technical Department	192.168.3.33	192.168.0.34	192.168.3.35	255.255.255.224	29
London HR Department	192.168.3.1	192.168.0.34	192.168.3.2	255.255.255.224	30

London Core VLAN Configuration:

```
London-Core#
London-Core#sh vlan b
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	London_HR_Department	active	
20	London_Technical_Department	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

London HR Side VLAN Configuration:

```
London-10#  
London-10#sh vlan b
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	London_HR_Department	active	Fa0/1, Fa0/2
20	London_Technical_Department	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

London Technical Side VLAN Configuration:

```
London-20#  
London-20#sh vlan b
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	London_HR_Department	active	
20	London_Technical_Department	active	Fa0/1, Fa0/2, Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Germany Office (Area 5)

Devices:

- 4 PCs (Germany Tech1, Germany Tech2, Germany HR1, Germany HR2)
- 1 server (Germany DHCP Server),
- 3 L-2 switches (Germany-Core, GerSVI-1, GerSVI-2)
- 1 Router (GER1)

Total Devices: 8

- **Departments:** HR (VLAN 10), Technical (VLAN 20).
- **IP Ranges:**
 - HR: 192.168.4.0/26 (Def Gateway: 192.168.4.1)
 - Technical: 192.168.4.32/26 (Def Gateway: 192.168.4.33)

Germany DHCP Configuration:

The screenshot shows a web-based configuration interface for a device named "Germany Server". The interface has a top navigation bar with tabs: Physical, Config, Services, Desktop (selected), Programming, and Attributes. Below the navigation bar, there is a blue header for "IP Configuration" with a close button (X). The main content area is titled "IP Configuration" and contains two radio buttons: "DHCP" (unselected) and "Static" (selected). Below the radio buttons, there are four input fields for static IP configuration:

Field	Value
IPv4 Address	192.168.4.34
Subnet Mask	255.255.255.224
Default Gateway	192.168.4.33
DNS Server	0.0.0.0

Germany Server

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

InterfaceFastEthernet0ServiceOnOff

Pool Name

serverPool

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

Start IP Address :

0000

Subnet Mask:

0000

Maximum Number of Users :

0

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User
serverPool	0.0.0.0	0.0.0.0	0.0.0.0	0.0.0.0	0
Germany Technical Department	192.168.4.33	192.168.0.34	192.168.4.35	255.255.255.224	29
Germany HR Department	192.168.4.1	192.168.0.34	192.168.4.2	255.255.255.224	30

Germany Core VLAN Configuration:

```
Germany-Core#
Germany-Core#sh vlan b
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	Germany_HR_Department	active	
20	Germany_Technical_Department	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Germany HR Side VLAN Configuration:

```
Germany-10#sh vlan b
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	Germany_HR_Department	active	Fa0/1, Fa0/2
20	Germany_Technical_Department	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Germany Technical Side VLAN Configuration:

```
Germany-20#  
Germany-20#sh vlan b
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Germany_HR_Department	active	
20	Germany_Technical_Department	active	Fa0/1, Fa0/2, Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Backbone Router (Area 0)

BBR1#

BBR1#show ip int b

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	administratively down	down
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/0/0	192.168.5.1	YES	manual	up	up
Serial0/0/1	192.168.5.5	YES	manual	up	up
Serial0/1/0	192.168.5.17	YES	manual	up	up
Serial0/1/1	192.168.5.13	YES	manual	up	up
Serial0/2/0	192.168.5.9	YES	manual	up	up
Serial0/2/1	unassigned	YES	unset	down	down
Serial0/3/0	192.168.5.21	YES	manual	up	up
Serial0/3/1	192.168.5.25	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

The Open Shortest Path First (OSPF) protocol is a link-state routing protocol designed for scalability and efficiency in IP networks. It operates by having routers exchange link-state advertisements to build a complete topology database, from which the shortest path to each destination is calculated using Dijkstra's algorithm. OSPF supports hierarchical network design through the use of areas, enhancing performance and manageability in large-scale deployments such as the multi-location network outlined in this project.

OSPF is superior to the Routing Information Protocol (RIP) for several reasons. RIP, a distance-vector protocol, relies on hop count as its metric, with a maximum limit of 15 hops, making it unsuitable for larger networks. It also converges slowly due to periodic updates every 30 seconds. In contrast, OSPF uses a cost metric based on link bandwidth, supports unlimited network scale through its area-based architecture, and achieves faster convergence by flooding link-state updates only when changes occur. For this project's multi-location network with five offices, OSPF's scalability, efficiency, and rapid convergence make it the preferred choice over RIP.

The area concept in OSPF is utilized to optimize network performance and scalability. By dividing a large network into smaller, manageable areas, OSPF reduces the volume of routing information exchanged between routers. Within an area, routers maintain a detailed link-state database, while only summarized routing information is shared with other areas via the backbone (Area 0). This minimizes processing overhead, reduces network traffic, and enhances stability. In this project, assigning each office (For e.g., Boston as Area 1, Mumbai as Area 2) to a separate area ensures efficient routing across geographically dispersed locations.

In OSPF, the backbone network is configured as Area 0 to serve as the central transit point for inter-area communication. All areas must connect to Area 0, either directly or via virtual links, to exchange routing information with other areas. This hierarchical design ensures a structured and stable routing architecture, preventing routing inefficiencies or isolation of areas. In this project, Area 0 acts as the backbone connecting the five office areas (Boston, Mumbai, New York, London, and Germany), facilitating efficient inter-office communication and maintaining network integrity.

OSPF relies on five distinct message types to maintain network operation:

- **Hello:** Sent periodically (default: 10 seconds) to discover and maintain neighbor relationships.
- **Database Description (DBD):** Exchanged during initial synchronization to summarize a router's link-state database.
- **Link-State Request (LSR):** Used by a router to request specific link-state updates from a neighbor.
- **Link-State Update (LSU):** Carries detailed LSAs to propagate topology changes or respond to LSRs.
- **Link-State Acknowledgment (LSAck):** Confirms receipt of LSUs, ensuring reliable transmission. These messages enable OSPF to establish adjacencies, synchronize databases, and update routing tables efficiently, as implemented across the project's five areas.

The security and redundancy plan for this network incorporates several measures:

- **Security:** Access Control Lists (ACLs) restrict Finance department traffic, allowing access only to other Finance departments and blocking all other inter-departmental access. SSH is configured on routers for secure remote management, and BPDU Guard prevents unauthorized switch additions. MAC flooding defense is implemented using port security to limit MAC addresses per port.

-
- **Redundancy:** HSRP is deployed in Boston and Mumbai offices with modified timers (Hello: 2 seconds, Hold: 6 seconds) for gateway redundancy. Rapid Spanning Tree Protocol (RSTP) with switch redundancy is implemented in Germany, London, and New York, ensuring rapid recovery from link failures. EtherChannel with LACP in New York enhances link reliability between switches.

The Spanning Tree Protocol (STP) prevents loops in Layer 2 networks by creating a loop-free logical topology. It operates as follows:

1. **Root Bridge Election:** Switches elect a root bridge based on the lowest Bridge ID (priority + MAC address).
2. **Path Cost Calculation:** Each switch calculates the least-cost path to the root bridge using link bandwidth-based costs.
3. **Port Roles Assignment:** Ports are assigned roles—Root Ports (best path to root), Designated Ports (forwarding traffic downstream), or Blocked Ports (preventing loops).
4. **BPDU Exchange:** Bridge Protocol Data Units (BPDUs) are sent every 2 seconds to share topology information and detect changes.
5. **Convergence:** STP transitions ports through states (Blocking, Listening, Learning, Forwarding) to stabilize the topology, taking 30–50 seconds traditionally. In this project, Rapid STP accelerates convergence, while PortFast and BPDU Guard enhance efficiency and security on host-connected ports. Loops are avoided by blocking redundant paths, ensuring a single active path per destination.

STP: The original protocol, it builds a single spanning tree for the entire network, preventing loops but underutilizing redundant links. Convergence takes 30–50 seconds, making it slower for modern networks.

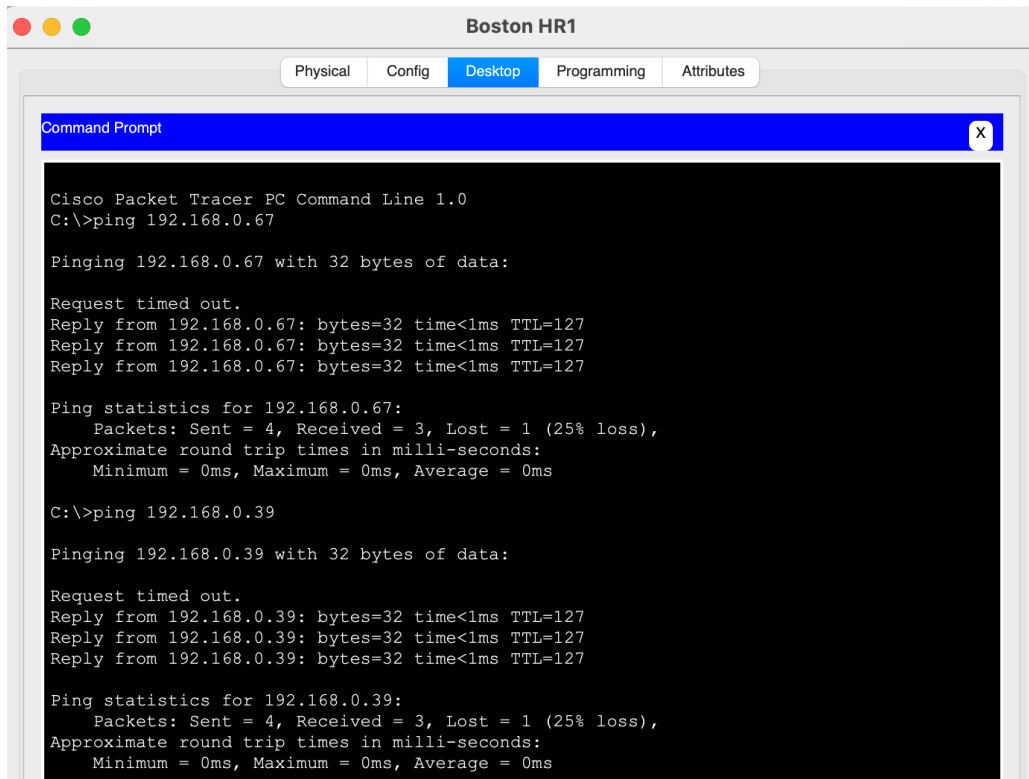
PVSTP (Per-VLAN Spanning Tree): A Cisco enhancement, PVSTP runs a separate STP instance per VLAN, allowing load balancing across redundant links by enabling different topologies per VLAN. It improves efficiency but increases resource usage with many VLANs.

MSTP (Multiple Spanning Tree Protocol): MSTP groups multiple VLANs into instances, reducing the number of spanning trees compared to PVSTP. It offers scalability and load balancing while optimizing resource usage, ideal for large networks. In this project, Rapid STP is used for Germany, London, and New York, balancing simplicity and faster convergence.

III. Test Plan for the Network

1. Test VLAN

Pinging different VLANs within the same subnet:



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.67

Pinging 192.168.0.67 with 32 bytes of data:

Request timed out.
Reply from 192.168.0.67: bytes=32 time<1ms TTL=127
Reply from 192.168.0.67: bytes=32 time<1ms TTL=127
Reply from 192.168.0.67: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.0.67:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.0.39

Pinging 192.168.0.39 with 32 bytes of data:

Request timed out.
Reply from 192.168.0.39: bytes=32 time<1ms TTL=127
Reply from 192.168.0.39: bytes=32 time<1ms TTL=127
Reply from 192.168.0.39: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.0.39:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping from Boston VLAN 10 PC to Boston VLAN 20 PC & VLAN 30 PC

Steps:

- Opened **Boston HR1's** Command Prompt.
- Pinged **Boston Fin1** and **Boston Tech1**.

Result: Successful pings

Inference: VLAN is Configured right

This step remains across all locations for testing VLANs.

2. Test OSPF

Scenario: Ping from Boston HR2 to London HR1

Show OSPF Neighbor (BOR1)

```
BOR1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
6.6.6.6	0	FULL/ -	00:00:30	192.168.5.1	Serial0/1/0
7.7.7.7	0	2WAY/DROTHER	00:00:31	192.168.0.38	GigabitEthernet0/0/0.20
7.7.7.7	0	2WAY/DROTHER	00:00:31	192.168.0.69	GigabitEthernet0/0/0.30
7.7.7.7	0	2WAY/DROTHER	00:00:31	192.168.0.5	GigabitEthernet0/0/0.10

Show OSPF Neighbor (Backbone Router)

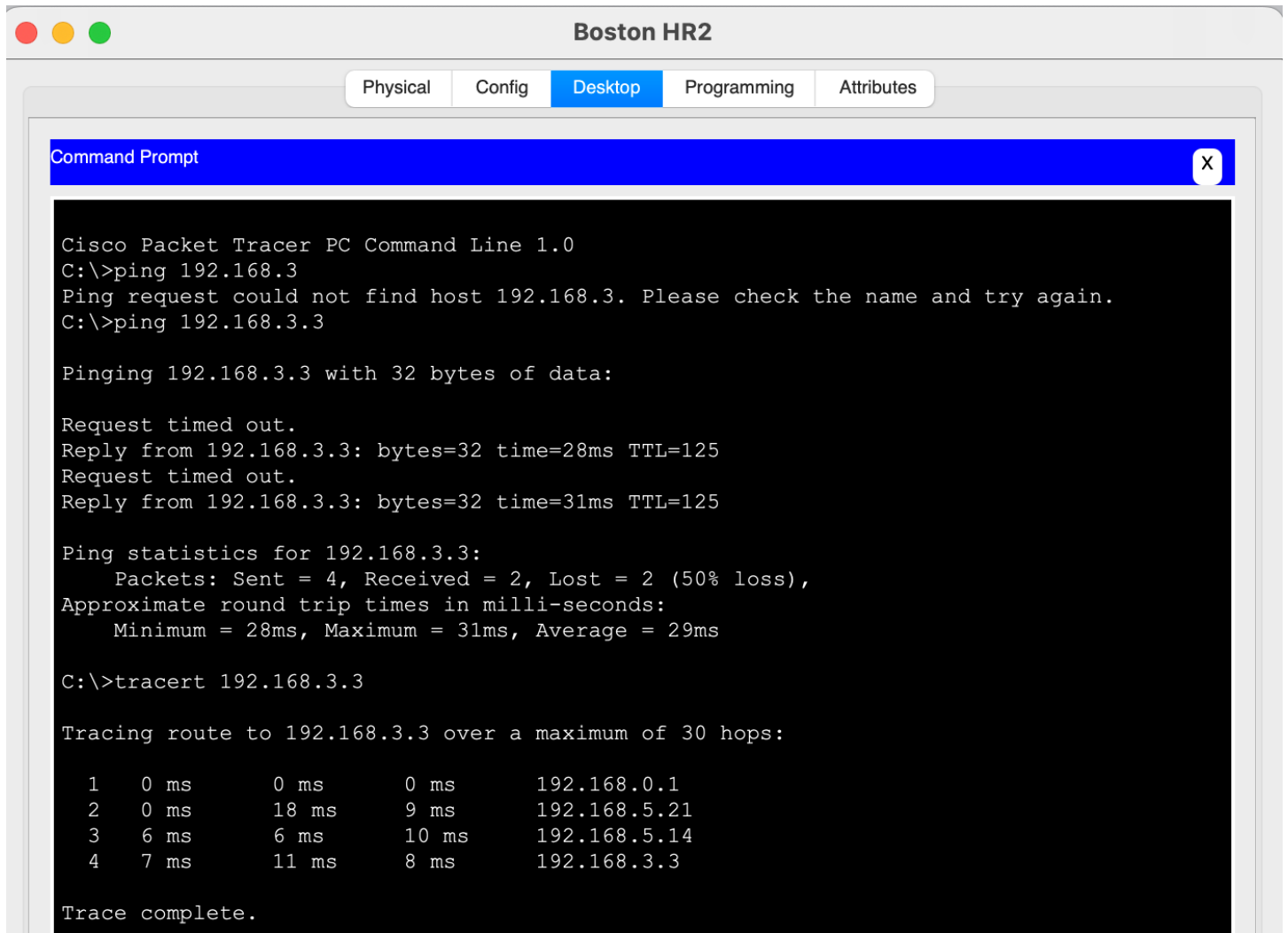
```
BBR1#show ip ospf n
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	0	FULL/ -	00:00:34	192.168.5.2	Serial0/0/0
2.2.2.2	0	FULL/ -	00:00:34	192.168.5.6	Serial0/0/1
5.5.5.5	0	FULL/ -	00:00:33	192.168.5.18	Serial0/1/0
4.4.4.4	0	FULL/ -	00:00:35	192.168.5.14	Serial0/1/1
3.3.3.3	0	FULL/ -	00:00:36	192.168.5.10	Serial0/2/0
7.7.7.7	0	FULL/ -	00:00:34	192.168.5.22	Serial0/3/0
8.8.8.8	0	FULL/ -	00:00:35	192.168.5.26	Serial0/3/1

Show OSPF Neighbor (London)

```
LOR1#show ip ospf n
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
6.6.6.6	0	FULL/ -	00:00:31	192.168.5.13	Serial0/1/0



Steps:

- Ran *show ip ospf neighbor* on Boston, Backbone, and London routers to verify OSPF adjacency.
- Pinged London HR1 from Boston HR2 to test connectivity.
- Used traceroute from Boston HR2 to London HR1 to map the path.

Results: Successful pings indicate correct OSPF routing configuration.

Inference: Routing is happening between two different regions, indicating the correct OSPF configuration in the network.

3. Test Security

```
C:\>ping 192.168.0.67

Pinging 192.168.0.67 with 32 bytes of data:

Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.

Ping statistics for 192.168.0.67:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Results & Inference:

The requirement was that the Finance department should not be accessible by any other departments, but Finance should have access to all other departments. Additionally, Finance departments should be able to communicate with each other.

I attempted to block inbound traffic to the Finance department in Boston, and I succeeded, as the ping results showed "Destination Host Unreachable." However, outbound pings from Finance also failed, which was not the intended outcome.

4. Test Redundancy Plan

Test HSRP

Boston Fin2

Physical Config **Desktop** Programming Attributes

Command Prompt

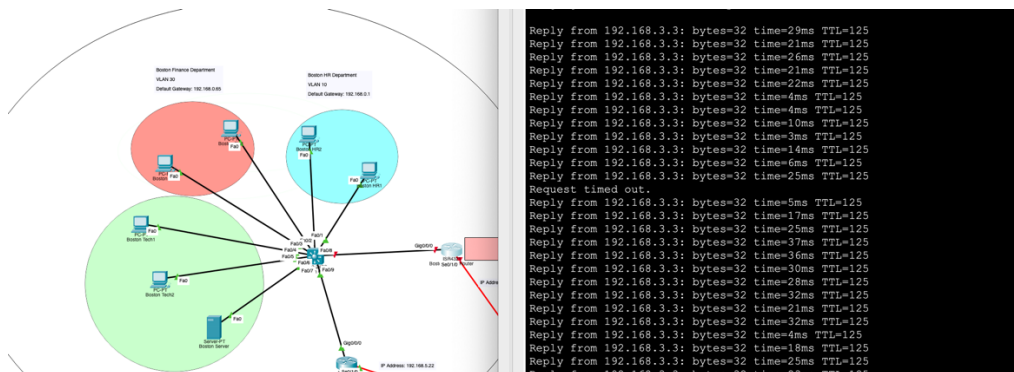
```
Request timed out.
Reply from 192.168.3.3: bytes=32 time=26ms TTL=125
Reply from 192.168.3.3: bytes=32 time=23ms TTL=125
Reply from 192.168.3.3: bytes=32 time=19ms TTL=125

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 19ms, Maximum = 26ms, Average = 22ms

C:\>ping -t 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Reply from 192.168.3.3: bytes=32 time=29ms TTL=125
Reply from 192.168.3.3: bytes=32 time=21ms TTL=125
Reply from 192.168.3.3: bytes=32 time=26ms TTL=125
Reply from 192.168.3.3: bytes=32 time=21ms TTL=125
Reply from 192.168.3.3: bytes=32 time=22ms TTL=125
Reply from 192.168.3.3: bytes=32 time=4ms TTL=125
Reply from 192.168.3.3: bytes=32 time=4ms TTL=125
Reply from 192.168.3.3: bytes=32 time=10ms TTL=125
Reply from 192.168.3.3: bytes=32 time=3ms TTL=125
Reply from 192.168.3.3: bytes=32 time=14ms TTL=125
Reply from 192.168.3.3: bytes=32 time=6ms TTL=125
Reply from 192.168.3.3: bytes=32 time=25ms TTL=125
Request timed out.
Reply from 192.168.3.3: bytes=32 time=5ms TTL=125
Reply from 192.168.3.3: bytes=32 time=17ms TTL=125
Reply from 192.168.3.3: bytes=32 time=25ms TTL=125
Reply from 192.168.3.3: bytes=32 time=37ms TTL=125
```



Steps:

- Pinged from Boston to London – successful.
- Turned off Boston's Active Router.
- Ping timed out briefly.
- Standby Router took over as Active.
- Ping resumed successfully.

Results:

- Temporary packet loss occurred.
- Connectivity restored after failover.

Inference:

- Standby router successfully took over, proving network redundancy and high availability.

The test is similar for verifying RSTP and switch redundancy as well.

5. Test MAC Flooding Attack

- Defend MAC Flooding Attack

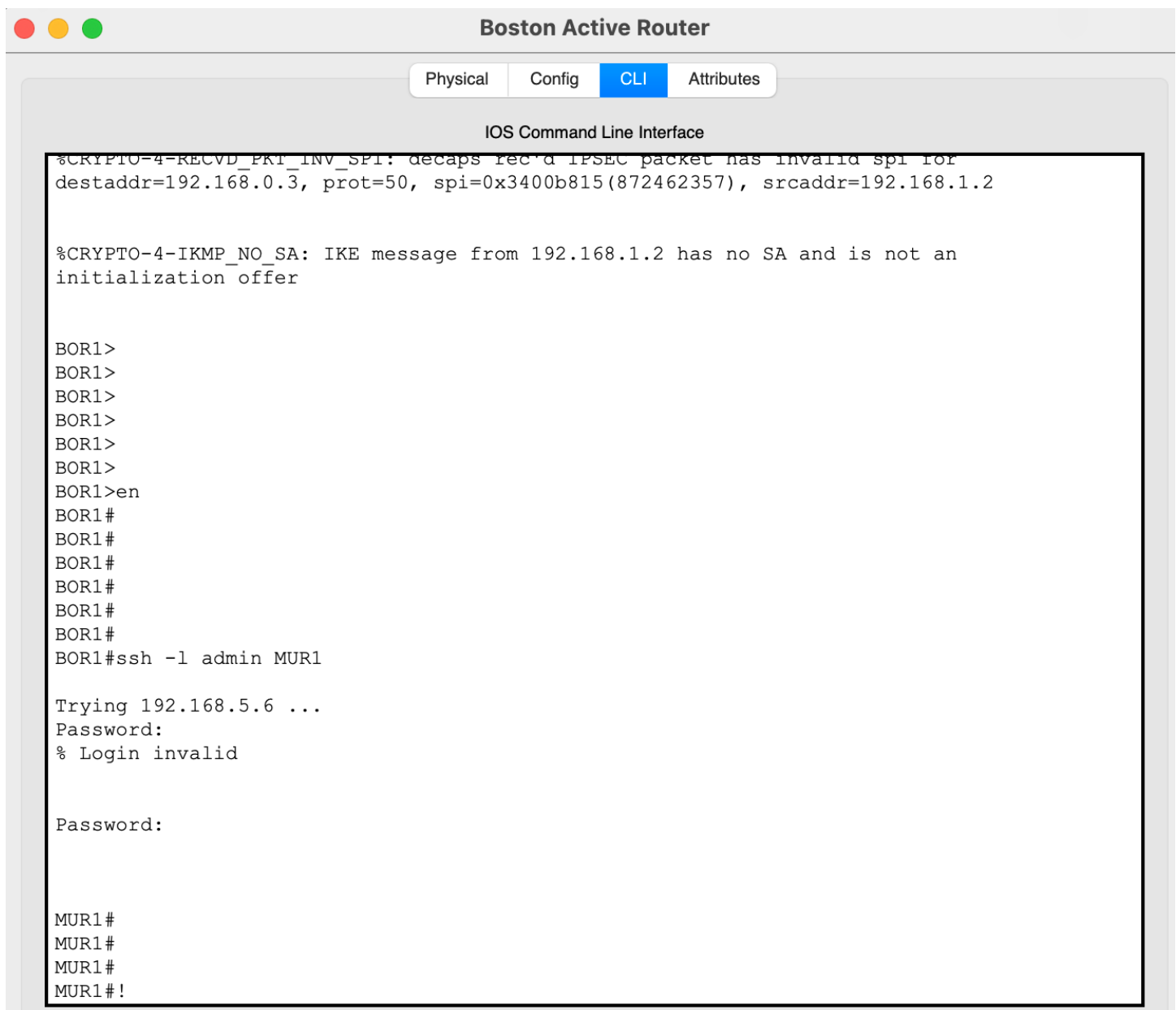
```
:
interface FastEthernet0/1
  switchport access vlan 10
  switchport mode access
  switchport port-security
  switchport port-security maximum 10
  switchport port-security violation restrict
!
interface FastEthernet0/2
  switchport access vlan 10
  switchport mode access
  switchport port-security
  switchport port-security maximum 10
  switchport port-security violation restrict
!
interface FastEthernet0/3
  switchport access vlan 30
  switchport mode access
  switchport port-security
  switchport port-security maximum 10
  switchport port-security violation restrict
!
interface FastEthernet0/4
  switchport access vlan 30
  switchport mode access
  switchport port-security
  switchport port-security maximum 10
  switchport port-security violation restrict
!
interface FastEthernet0/5
  switchport access vlan 20
  switchport mode access
  switchport port-security
  switchport port-security maximum 10
  switchport port-security violation restrict
!
interface FastEthernet0/6
  switchport access vlan 20
  switchport mode access
  switchport port-security
  switchport port-security maximum 10
  switchport port-security violation restrict
!
interface FastEthernet0/7
  switchport access vlan 20
  switchport mode access
  switchport port-security
  switchport port-security maximum 10
  switchport port-security violation restrict
,
```

Steps:

Port Security Enabled: Each interface (FastEthernet0/1 to FastEthernet0/7) has the following commands:

- **switchport port-security:** This enables port security on the interface.
- **switchport port-security maximum 10:** This limits the number of MAC addresses that can be learned on the port to 10.
- **switchport port-security violation restrict:** This specifies that if the maximum number of MAC addresses is exceeded (i.e., a potential MAC flooding attack), the switch will restrict the port by dropping packets from additional MAC addresses while keeping the port operational (as opposed to shutting it down or sending an SNMP trap).

- SSH using hostnames of the Routers:



The screenshot shows a web-based interface for a "Boston Active Router". At the top, there are four tabs: "Physical", "Config", "CLI" (which is selected and highlighted in blue), and "Attributes". Below the tabs, the text "IOS Command Line Interface" is displayed. The main area contains a terminal window with the following text:

```
%CRYPTO-4=RECV_PRT_INV_SPI: decaps rec'd IPSEC packet has invalid spi for
destaddr=192.168.0.3, prot=50, spi=0x3400b815(872462357), srcaddr=192.168.1.2

%CRYPTO-4-IKMP_NO_SA: IKE message from 192.168.1.2 has no SA and is not an
initialization offer

BOR1>
BOR1>
BOR1>
BOR1>
BOR1>
BOR1>
BOR1>
BOR1>en
BOR1#
BOR1#
BOR1#
BOR1#
BOR1#
BOR1#
BOR1#ssh -l admin MUR1

Trying 192.168.5.6 ...
Password:
% Login invalid

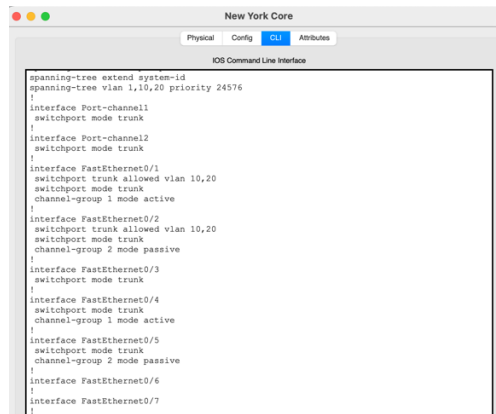
Password:

MUR1#
MUR1#
MUR1#
MUR1#!
```

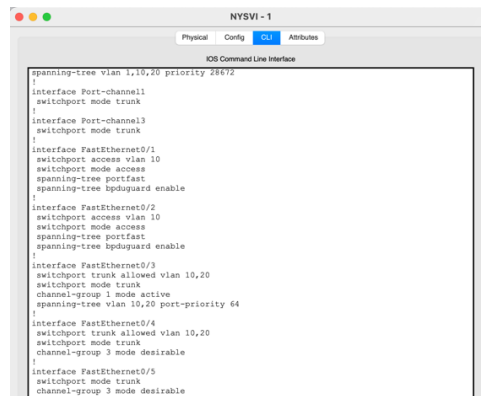
Steps:

- **Initiated an SSH connection:** Used the command `ssh -l admin MUR1` to connect to the router at IP address 192.168.5.6 with the username "admin".
- **Attempted authentication:** Entered a password when prompted by the router to authenticate the SSH session.
- Upon successful login, the CLI would change to MUR1>.

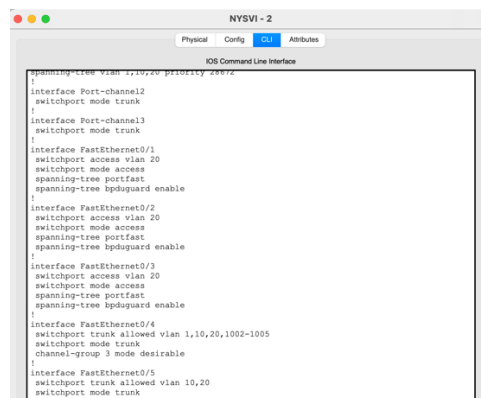
• EtherChannel Configuration with LACP on NY



```
spanning-tree extend system-id
spanning-tree vlan 1,10,20 priority 24576
!
interface Port-channel1
switchport mode trunk
!
interface Port-channel2
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk allowed vlan 10,20
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/2
switchport trunk allowed vlan 10,20
switchport mode trunk
channel-group 2 mode passive
!
interface FastEthernet0/3
switchport mode trunk
!
interface FastEthernet0/4
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/5
switchport mode trunk
channel-group 2 mode passive
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
```



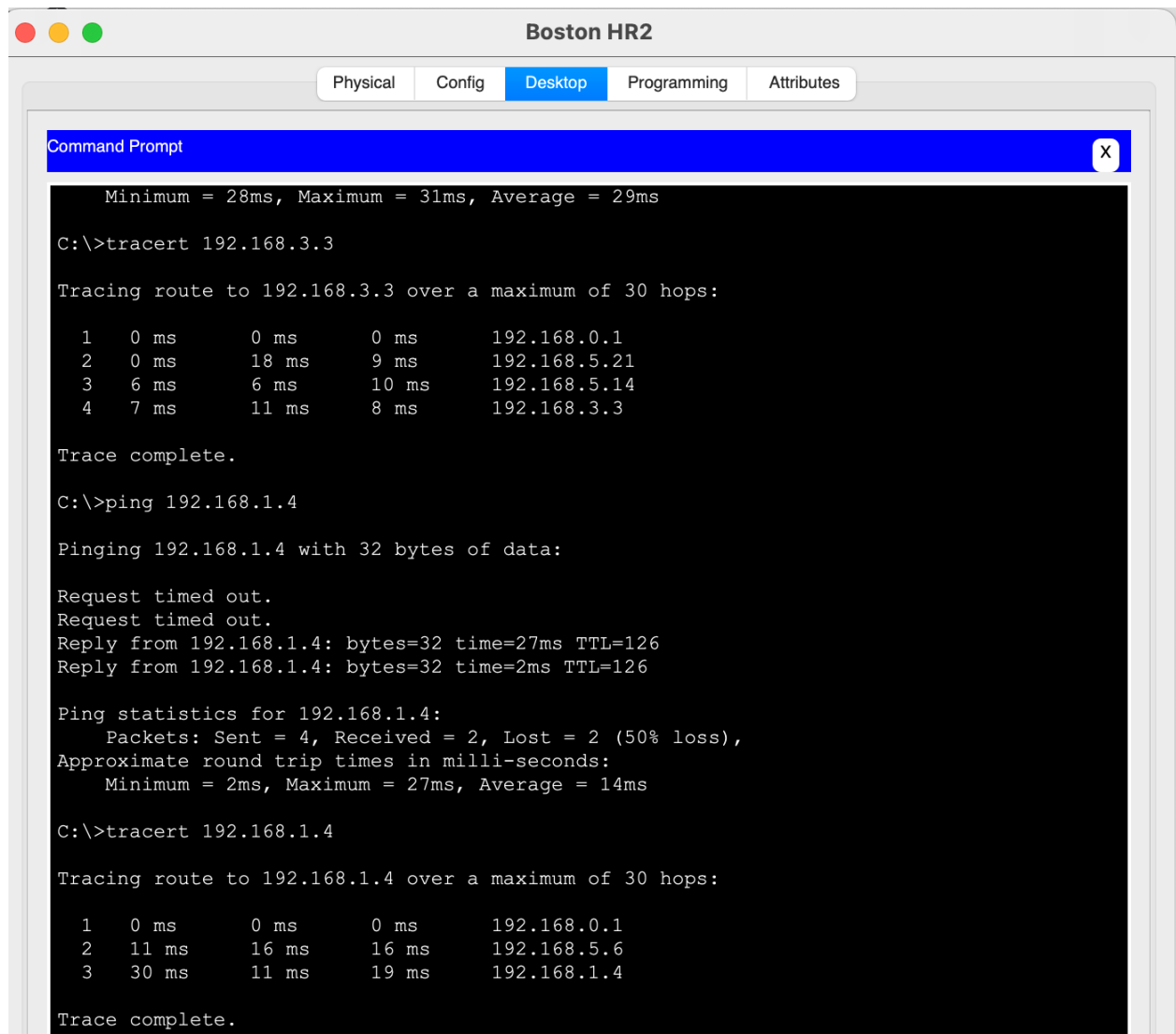
```
spanning-tree vlan 1,10,20 priority 28672
!
interface Port-channel1
switchport mode trunk
!
interface Port-channel3
switchport mode trunk
!
interface FastEthernet0/1
switchport access vlan 10
switchport mode access
spanning-tree portfast
spanning-tree bpdguard enable
!
interface FastEthernet0/2
switchport access vlan 10
switchport mode access
spanning-tree portfast
spanning-tree bpdguard enable
!
interface FastEthernet0/3
switchport trunk allowed vlan 10,20
switchport mode trunk
channel-group 1 mode active
spanning-tree vlan 10,20 port-priority 64
!
interface FastEthernet0/4
switchport trunk allowed vlan 10,20
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/5
switchport mode trunk
channel-group 3 mode desirable
!
```



```
spanning-tree vlan 1,10,20 priority 28672
!
interface Port-channel2
switchport mode trunk
!
interface Port-channel3
switchport mode trunk
!
interface FastEthernet0/1
switchport access vlan 20
switchport mode access
spanning-tree portfast
spanning-tree bpdguard enable
!
interface FastEthernet0/2
switchport access vlan 20
switchport mode access
spanning-tree portfast
spanning-tree bpdguard enable
!
interface FastEthernet0/3
switchport access vlan 20
switchport mode access
spanning-tree portfast
spanning-tree bpdguard enable
!
interface FastEthernet0/4
switchport trunk allowed vlan 1,10,20,1002-1005
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/5
switchport trunk allowed vlan 10,20
switchport mode trunk
!
```

EtherChannel with LACP Configuration: Interfaces on NY-SVI-1, NY-SVI-2, and New York Core are bundled into EtherChannel groups (e.g., Port-channel1, Port-channel2) using LACP in active, passive, or desirable modes to aggregate FastEthernet links, enabling load balancing and redundancy for VLANs 1, 10, and 20 traffic.

• VPN IPsec Tunnel between Boston and Mumbai



The screenshot shows a Windows desktop window titled "Boston HR2" with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window. The Command Prompt shows the results of a traceroute to 192.168.3.3 and a ping to 192.168.1.4.

```
Minimum = 28ms, Maximum = 31ms, Average = 29ms

C:\>tracert 192.168.3.3

Tracing route to 192.168.3.3 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.0.1
  2  0 ms    18 ms   9 ms    192.168.5.21
  3  6 ms    6 ms    10 ms   192.168.5.14
  4  7 ms    11 ms   8 ms    192.168.3.3

Trace complete.

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.1.4: bytes=32 time=27ms TTL=126
Reply from 192.168.1.4: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 27ms, Average = 14ms

C:\>tracert 192.168.1.4

Tracing route to 192.168.1.4 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.0.1
  2  11 ms   16 ms   16 ms   192.168.5.6
  3  30 ms   11 ms   19 ms   192.168.1.4

Trace complete.
```

Steps:

1. Pinged from Boston to London – took 4 hops, passing through BBR1.
2. Pinged from Boston to Mumbai – took 3 hops, skipping BBR1.
3. Observed that the ping to Mumbai bypassed BBR1, suggesting an alternative path.

Results:

- **Boston to London:** 4 hops, via BBR1.
- **Boston to Mumbai:** 3 hops, bypassing BBR1.

Inference:

- The 3-hop route to Mumbai indicates the presence of a **tunnel** between Boston and Mumbai, bypassing the usual path through BBR1.
- The IPsec tunnel was successfully established, providing a direct communication link between the two locations.

- EIGRP and Loopback advertisement inside EIGRP

```
!
router eigrp 1
  eigrp router-id 1.1.1.1
  network 192.168.6.0 0.0.0.31
  network 192.168.0.0 0.0.0.31
  network 192.168.0.32 0.0.0.31
  network 192.168.0.64 0.0.0.31
  network 1.1.1.1 0.0.0.0
!
```

```
!
router eigrp 1
  eigrp router-id 2.2.2.2
  network 192.168.6.0 0.0.0.31
  network 192.168.0.0 0.0.0.31
  network 192.168.0.32 0.0.0.31
  network 192.168.0.64 0.0.0.31
  network 2.2.2.2 0.0.0.0
!
```

```
!
interface Loopback0
  ip address 1.1.1.1 255.255.255.255
!
```

```
!
interface Loopback0
  ip address 2.2.2.2 255.255.255.255
!
```

Steps:

- **EIGRP Configuration:** Both routers are configured with EIGRP AS 1, with Router 1 (EIGRP router-id 1.1.1.1) and Router 2 (EIGRP router-id 2.2.2.2) advertising networks in the 192.168.0.0/16 range and their respective loopback addresses (1.1.1.1 and 2.2.2.2).
- **Loopback Interfaces:** Router 1 has Loopback0 with IP 1.1.1.1/32, and Router 2 has Loopback0 with IP 2.2.2.2/32, used for stable router identification in EIGRP.
- **Network Advertisement:** The routers advertise their local networks (192.168.0.0/19, 192.168.0.32/19, 192.168.0.64/19) and loopback addresses to establish EIGRP neighbor relationships and share routing information.

IV. Concepts and Tools Learned from the Project Work

Concepts

1. IP Addressing and Subnetting
2. Network Design and Architecture
3. DHCP Configuration
4. Routing Protocols (OSPF and EIGRP)
5. VLANs and Trunking
6. Rapid STP, PortFast, and BPDU Guard
7. Security Mechanisms
8. Redundancy Techniques (HSRP)
9. VPN Configuration
10. DNS Setup
11. Network Optimization
12. Testing and Validation

Tools or Devices

1. Cisco Packet Tracer
2. Routers (e.g., ISR4321)
3. Switches
4. DHCP Servers
5. DNS Server
6. Access Points
7. Workstations (PCs)

V. Conclusion:

This project delivered a multi-location network for an organization across Boston, Mumbai, New York, London, and Germany. By subnetting 192.168.0.0/16, implementing OSPF with distinct areas, and integrating HSRP, VLANs, and a VPN, the design met all requirements for scalability, security, and redundancy. This project helped me sharpen my skills in IP addressing, routing protocols, DHCP, and network optimization while emphasizing the importance of testing and security (e.g., ACLs, MAC flooding defense). It bridged theory and practice, preparing me for real-world networking challenges with a solid, efficient solution.